

Testing Additional Gravitational Degrees of Freedom with the Th-229 Nuclear Clock

An experimental design (Route B: verify first, engineer later)

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Abstract The goal of this design is not an “anti-gravity prototype” but a test of whether gravity possesses controllable additional degrees of freedom. The Th-229 nuclear clock is the core instrument, measuring the gravitational modulation of the isomer transition frequency on an elliptical orbit. Classical general relativity fixes the baseline (a fractional modulation of 1.3×10^{-11}); a Kaluza-Klein extra dimension superimposes $\Delta E \approx 2.8\alpha$ eV, where α is the coupling of the lightest extra mode. At the target fractional stability of 10^{-18} , the experiment reaches α down to 3×10^{-18} for ranges $\lambda > 10^7$ m, about three orders beyond current orbital bounds. The design proceeds in two phases: Phase 1 establishes the classical baseline and eliminates systematics; Phase 2 searches for the deviation and pins or opens the window. An advanced phase actively controls the nuclear spin state to test whether the coupling is state-dependent. This document provides the parameter tables, expected signal magnitudes, systematic-error budget, and verdict formulas. A null result carries publication value by itself: $\alpha < 10^{-17}$ is a PRL-grade constraint. **Keywords** Th-229; nuclear clock; Kaluza-Klein; gravitational Aharonov-Bohm; equivalence principle; precision metrology

Scientific positioning and goals

Positioning: this experiment belongs to the established frontier of “testing the fundamental properties of gravity and exploring their controllability”; it is not the development of an “anti-gravity device”. Expression discipline: in review and defense settings, use “tests of additional gravitational degrees of freedom”, never “anti-gravity” as the experimental goal.

Goals, layered:

- **Goal 1 (mandatory):** the first nuclear-transition measurement of the gravitational redshift / gravitational Aharonov-Bohm effect, establishing the classical baseline.
- **Goal 2 (mandatory):** a constraint or a signal on α in the neV window, pinning or opening the Kaluza-Klein parameter space.
- **Goal 3 (optional, advanced):** active control of the nuclear spin state, testing whether the gravitational coupling is state-dependent. A positive result would simultaneously falsify general relativity and standard Kaluza-Klein, pointing to a new coupling channel.

Null-result value: a null result for Goal 2 yields $\alpha < 10^{-17}$ -level constraints, two orders beyond current orbital bounds, independently publishable at PRL / Nature Physics level.

Physics and window position

Signal form

The lightest graviton mode of an extra compact dimension contributes a Yukawa correction

$$V(r) = -\frac{Gm_1m_2}{r}(1 + \alpha e^{-\frac{r}{\lambda}}), \quad (1)$$

with α the coupling relative to standard gravity and λ the Compton range. A quantum system in free fall around a gravitating body acquires a splitting

$$\Delta E = m_{\text{sys}}\alpha\Delta\Phi(r, \lambda), \quad (2)$$

with $\Delta\Phi$ the peak-to-peak potential variation. For the thorium nucleus ($m_{\text{Th}} \approx 3.8 \times 10^{-25}$ kg) on a low Earth orbit (altitude 400 km, eccentricity 0.01), $\Delta\Phi \approx 2gae \approx 1.18 \times 10^6 \frac{\text{m}^2}{\text{s}^2}$ (classical part), hence

$$\Delta E \approx 2.8\alpha \text{ eV}. \quad (3)$$

Window position

Inverting Eq. Equation 2 against history: electronvolt splittings for nuclear systems require $\alpha \approx 0.4$, excluded by seven to eight orders of magnitude; millielectronvolt splittings require $\alpha \approx 10^{-3}$, equally excluded. The allowed window therefore sits at neV and below. Eq. Equation 3 shows that the current orbital bound $\alpha \sim 10^{-15}$ corresponds to $\Delta E \sim 3$ feV, precisely inside the reach of the target nuclear-clock sensitivity. That is the window position: **not “signal or no signal”, but “who fills the three orders of magnitude between the existing bound and the clock sensitivity first”**.

Classical baseline

The classical gravitational-redshift modulation is the peak-to-peak orbital potential variation:

$$\frac{\Delta\nu}{\nu} = \frac{\Delta\Phi}{c^2} \approx 1.31 \times 10^{-11}, \quad (4)$$

at the orbital frequency $f_{\text{orb}} \approx 1.80 \times 10^{-4}$ Hz. Numerically this baseline equals the classical redshift variation (the reading of the gravitational Aharonov-Bohm effect); it serves three purposes: a systematics target, a general-relativity test with a nuclear clock, and the reference for Phase-2 signal subtraction.

Experimental design: two phases plus an advanced stage

Phase 1: classical baseline

- **Instrument:** solid-state Th-229 nuclear clock (148.18 nm direct excitation achieved; CaF₂ host with 447 ± 25 s half-life; X-ray quenching readout accelerating cycles by $\geq 50\times$).
- **Operation:** place the clock in a known gravitational-potential difference (ground: 1 m height, $g\Delta\frac{h}{c^2} \approx 1.09 \times 10^{-16}$; orbit: the full modulation of Eq. Equation 4), and measure the isomer frequency shift, reproducing the general-relativistic prediction.
- **Verdict:** the ground baseline requires fractional stability better than 10^{-16} (the spinless-host route, Th(SO₄)₂, projects $4.6 \times \frac{10^{-23}}{\sqrt{\tau}}$); the orbital baseline is four orders less demanding and can serve early systematics commissioning.

Phase 2: the Kaluza-Klein deviation search

- **Operation:** on orbit, fit the total modulation amplitude $A_{\text{tot}} = A_{\text{GR}} + A_{\text{KK}}$ at the orbital frequency, with A_{GR} modeled exactly by Eq. Equation 4; the residual is $A_{\text{KK}} = 2.8\alpha$ eV.
- **Null verdict:** $|A_{\text{KK}}| < 3\sigma_{\text{clock}}$ gives the constraint curve $\alpha < \alpha_{\min(\lambda)}$ (Table 3, Section 5).
- **Signal verdict:** a nonzero residual that is (i) neV-scale, (ii) phase-consistent with the Yukawa correction, and (iii) independent of the classical model (cross-checked across different altitudes or eccentricities).

Advanced stage: active-control demonstration

- **Operation:** apply microwave/magnetic fields to control the nuclear spin polarization and observe whether the coupling varies with the control parameters.
- **Theoretical positioning (mandatory):** in standard Kaluza-Klein the coupling α is a constant and does **not** depend on the nuclear spin state; a measured state dependence would falsify not only general relativity but standard Kaluza-Klein, pointing to a new, state-dependent gravitational coupling channel. The correct statement of this stage is therefore “search for state-dependent gravitational coupling”, not “tuning α ”. This discipline is what separates the design from folk proposals.

Core parameter tables

Instrument parameters (Table 1)

Parameter	Value	Source/status
Transition energy	8.355733554021(8) eV	2024 precision measurement
Wavelength / half-life	148.18 $\pm$ 0.42 nm; 447 $\pm$ 25 s (CaF ₂)	Nat. Commun. 2024
Solid-state lifetime	641 s (CaF ₂)	2025 review
Current linewidth	kHz-scale (crystal broadening)	present
Target stability	$4.6 \times \frac{10^{-23}}{\sqrt{\tau}}$ (Th(SO ₄) ₂)	theory, 2025
Readout	X-ray quenching, cycle speedup $\geq 50\times$	Nat. Commun. 2024

Table 1: Table 1: Th-229 nuclear clock instrument parameters.

Parameter	Value	Note
Orbit	400 km altitude, $e = 0.01$	LEO, $a = 6771$ km
Orbital frequency	1.80×10^{-4} Hz	the modulation frequency
Classical $\Delta\Phi$	1.18×10^6 m ² /s ²	peak-to-peak, $2gae$
Classical $\Delta\frac{v}{v}$	1.31×10^{-11}	Eq. (4)
Nucleus mass	3.8×10^{-25} kg	m_{Th}
Signal coefficient	$\Delta E = 2.8\alpha$ eV	Eq. (3)

Table 2: Table 2: Orbital platform and signal coefficient.

Range λ	Suppression $e^{-\frac{\alpha}{\lambda}}$	$\alpha_{\min}$ (10^{-18} target)	Existing bound
10^6 m	1.1×10^{-3}	2.7×10^{-15}	$\sim 10^{-15}$ (MICROSCOPE)
3×10^6 m	0.10	2.9×10^{-17}	$\sim 10^{-15}$
10^7 m	0.51	5.9×10^{-18}	$\sim 10^{-15}$
10^9 m	0.99	3.0×10^{-18}	$\sim 10^{-11}$ (LLR)

Table 3: Table 3: Reachability versus existing bounds. $\alpha_{\min} = \frac{\delta E}{m_{\text{Th}} \Delta\Phi} \times e^{\frac{\alpha}{\lambda}}$, with $\delta E = 8.36$ eV $\times 10^{-18}$.

The experiment deepens existing bounds by 2-8 orders of magnitude in the $\lambda > 10^6$ m window.

Platform parameters (Table 2)

Reachability (Table 3)

Systematic-error budget

Reading: the main systematic battlefield is not sensitivity but **the exact modeling of the classical baseline**: the Phase-2 signal shares the orbital frequency with the classical modulation and must be separated by modeling precision, not by spectral discrimination. This is structurally identical to the gravitational Aharonov-Bohm interpretation, and it is the reason Phase 1 exists.

Verdict formulas

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- **Null verdict:** $|A_{\text{KK}}| < 3\sigma$ implies $\alpha < \alpha_{\min}(\lambda)$, with $\alpha_{\min}$ from Table 3; publication form is the constraint curve $\alpha(\lambda)$.

Source	Scale (present)	Control	Target
Orbit determination	cm-level $\rightarrow \delta \frac{\Phi}{c^2} \sim 10^{-15}$	SLR/GNSS precise orbit determination + reprocessing	mm-level $\rightarrow 10^{-16}$
Gravity-field model	J2+ multipoles $\sim 10^{-10}$	high-order multipoles + time-variable models	10^{-18} residuals
Tides / atmosphere	time-variable, $\sim 10^{-12}$	common-mode subtraction + dual clocks	$\leq 10^{-17}$
Crystal-field quadrupole	kHz-scale (current dominant)	spinless host + quadrupole suppression	mHz-scale
Temperature drift	parametric (to be calibrated)	mK temperature control + in-situ calibration	group-calibrated
Zeeman	first-order sensitive	magic-angle / spinless host	first-order eliminated
X-ray quenching cycle	duty-cycle effects	cycle phase locking	$\leq 10^{-18}$
Space-ground link	$\sim 10^{-16}$	frequency-comb / two-way microwave link	10^{-18}

Table 4: Table 4: Systematic-error budget. Entries marked “parametric” require calibration by the host group; the table gives control methods and target magnitudes, not promises.

Milestone	Content	Prerequisite	Output
M1	Parameter space, reachability, error budget (this document)	none	theory paper (submittable now)
M2	Ground nuclear-clock redshift test (1 m, 1.09×10^{-16})	spinless host at 10^{-16}	GR test + systematics commissioning
M3	Orbital baseline + KK search	M2 + flight opportunity	$\alpha < 10^{-17}$ or a signal
M4	Active-control demonstration	M3 positive signal	state-dependence verdict

Table 5: Table 5: Milestones. M1 is achievable now; M2-M4 depend on host-material and flight-opportunity progress.

- **Signal verdict:** a nonzero residual satisfying (i) neV magnitude, (ii) Yukawa phase behavior, (iii) cross-validation across orbit configurations.
- **State-dependence verdict (advanced):** coupling varying with nuclear spin polarization falsifies GR and standard KK and points to a new channel; invariance excludes state-dependent coupling down to the experimental sensitivity.

Milestones

Collaboration interface

- **Theory collaborator provides:** signal predictions (Eqs. 1-4), the reachability table (Table 3), the error budget (Table 4), the verdict formulas (Section 7), parameter-scan code, and the three-model discrimination simulations (all in the companion repository).
- **Host group provides:** the clock platform, in-situ calibration of environmental shifts, and an orbital or ground-based potential-difference platform.

- **Candidate groups:** Huazhong University of Science and Technology gravity center, University of Science and Technology of China precision-measurement institutes, SIOM, and the European Th-229 clock groups.
- **Cost model:** the theory collaborator carries zero hardware cost; the ground baseline can run on existing clock platforms.

Compatibility with existing constraints

Every parameter of this design lies in the unexcluded window: the target region ($\lambda > 10^6$ m, $\alpha > 10^{-18}$) sits in the open region below the MICROSCOPE ($\alpha \sim 10^{-15}$) and LLR bounds; the experiment introduces no new interaction hypothesis (standard KK is an existing framework); it conflicts with neither Eöt-Wash nor short-range fifth-force bounds. The design introduces exactly one conjecture, and it is explicitly labeled (Section 3.3: state-dependent coupling is a **hypothesis to be tested**, not a premise of the design).

Conclusion

This design answers the highest-value question at the lowest cost: **does spacetime possess controllable additional degrees of freedom?** The two-phase structure guarantees an independent output at every step: the baseline alone is the first nuclear-clock gravitational redshift test; the null result of the deviation search is a two-order constraint improvement; a positive signal opens the first door to gravitational engineering. The academic safety of the design rests on three pillars: goal-statement discipline (test properties, claim no engineering), null-result value (a constraint is a result), and parameter compliance (everything inside the unexcluded window).

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